Computability & Logic, Exercises to Lectures 23+24

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Theoretical Exercises

From Huth and Ryan book

Exercises: Predicate Logic

1. 2.5.1 (a, c, e). Recall that a model $\mathcal{M}$ is defined over a set of elements, e.g. $A = \{a, b, c\}$, and some relations on these, e.g. $P^{\mathcal{M}} = \{(a, b)\}$ and $Q^{\mathcal{M}} = \{(a), (c)\}$, here meaning that exactly $P(a, b)$, $Q(a)$, and $Q(c)$ are true. Note that for relations with arity zero (such as L in part c), the model sets L to always be either true or false.

2. 2.5.2

3. 2.3.1: prove (a, b) in LogicBox.

4. 2.3.2

5. 2.3.3 (b, then a)

6. 2.4.12 (k): Prove $(\forall x.\exists y.(P(x) \to Q(y))) \to (\exists y.\forall x.(P(x) \to Q(y)))$ in LogicBox.

Exercises: Peano Arithmetic

7. Prove the following in Peano Arithmetic:

   a. $\forall x, y, z.(x + y) + z = x + (y + z)$. (Optional: use LogicBox)

   b. $\forall x, y.x + y = y + x$. (Optional: use LogicBox). Hint: Use these for free: $\forall x.x + 0 = 0 + x$ and $\forall x.x + 1 = 1 + x$. If you want an extra challenge, also prove these.

8. Consider the AR model of arithmetic, with $\mathbb{Z}$ instead of $\mathbb{N}$. Which of the axioms of Peano Arithmetic fail?

9. The macro isPrime(x) is true exactly when x is prime, and $x|n$ is true exactly when x is a divisor of n. Consider the following three predicates:

$$\begin{aligned}
\mathrm{Foo}_1(n) &:= \forall x.\mathrm{isPrime}(x) \land x|n \to x = 2 \\
\mathrm{Foo}_2(n) &:= \forall x.\mathrm{isPrime}(x) \land x|n \to (x = 2 \lor x = 3) \\
\mathrm{Foo}_3(n) &:= (\forall x.\mathrm{isPrime}(x) \land x|n \to x = 2) \lor (\forall x.\mathrm{isPrime}(x) \land x|n \to x = 3)
\end{aligned}$$

   a. Check for each of $\mathrm{AR} \models \mathrm{Foo}_1(1)$, $\mathrm{AR} \models \mathrm{Foo}_1(2)$, ..., $\mathrm{AR} \models \mathrm{Foo}_1(10)$ whether they hold. Can you characterize $\mathrm{Foo}_1(n)$ generally?

   b. Check for each of $\mathrm{AR} \models \mathrm{Foo}_2(1)$, $\mathrm{AR} \models \mathrm{Foo}_2(2)$, ..., $\mathrm{AR} \models \mathrm{Foo}_2(10)$ whether they hold. Can you characterize $\mathrm{Foo}_2(n)$ generally?

   c. Find n s.t. $\mathrm{AR} \models \mathrm{Foo}_2(n)$ but $\mathrm{AR} \not\models \mathrm{Foo}_3(n)$.

Bonus Exercises

10. Prove the following in Peano Arithmetic:

    a. $\forall x, y, z.(x * y) * z = x * (y * z)$. (Optional: use LogicBox)

    b. $\forall x, y.x * y = y * x$. (Optional: use LogicBox)

11. Consider the AR model of arithmetic, with $\mathbb{Z} \mod 2$ instead of $\mathbb{N}$. Which of the axioms of Peano Arithmetic fail?

12. 2.5.1 (b, d, f, g)

Reference: Axioms of Peano Arithmetic

1. $\forall x.x + 0 = x$
2. $\forall x \forall y.x + (y + 1) = (x + y) + 1$
3. $\forall x.x * 0 = 0$
4. $\forall x \forall y.x * (y + 1) = x * y + x$
5. $\forall x.\neg(0 = x + 1)$
6. $\forall x \forall y.(x + 1 = y + 1 \rightarrow x = y)$
7. $\varphi[0] \wedge (\forall x.\varphi[x] \rightarrow \varphi[x + 1]) \rightarrow \forall x.\varphi[x]$